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The impact of obesity on reproductive health and metabolism in reproductive-age females

Samantha B. Schon, M.D., M.T.R.^a, Hannah E. Cabre, Ph.D., R.D.N.^b, Leanne M. Redman, Ph.D.^b

^aDivision of Reproductive Endocrinology, Department of Obstetrics and Gynecology, University of Michigan, Ann Arbor, Michigan

^bReproductive Endocrinology and Women's Health Laboratory, Pennington Biomedical Research Center, Baton Rouge, Louisiana

Abstract

Obesity is a highly prevalent chronic disease that impacts >40% of reproductive-aged females. The pathophysiology of obesity is complex and can be understood simply as a chronic energy imbalance whereby caloric intake exceeds caloric expenditure with an energy surplus stored in adipose tissue. Obesity may be categorized into degrees of severity as well as different phenotypes on the basis of metabolic health and underlying pathophysiology. Obesity and excess adiposity have a significant impact on fertility and reproductive health, with direct effects on the hypothalamic-pituitary-ovarian axis, the ovary and oocyte, and the endometrium. There are significant adverse pregnancy outcomes related to obesity, and excess weight gain before, during, and after pregnancy that can alter the lifelong risk for metabolically unhealthy obesity. Given the high prevalence and pervasive impact of obesity on reproductive health, there is a need for better and individualized care for reproductive-aged females that considers obesity phenotype, underlying pathophysiology, and effective and sustainable interventions to treat obesity and manage weight gain before, during, and after pregnancy.

Keywords

Obesity; adipokines; infertility; reproductive health; fertility treatment

EPIDEMIOLOGY OF OBESITY

Excess body fat mass, or obesity, is the most prevalent noncommunicable disease worldwide. The World Health Organization defines obesity as abnormal or excessive fat accumulation that introduces undue risks to health (1). Body fat status is classified clinically using the body mass index (BMI)—a ratio of weight to height²—a value of ≥ 30 kg/m² is

Correspondence: Samantha B. Schon, M.D., M.T.R., Division of Reproductive Endocrinology, Department of Obstetrics and Gynecology, University of Michigan, L4000 University Hospital South, 1500 E. Medical Center Drive Ann Arbor, Michigan 48109 (sschon@med.umich.edu).

CRedit Authorship Contribution Statement

Samantha B. Schon: Writing – review & editing, Writing – original draft, Conceptualization. **Hannah E. Cabre:** Writing – original draft. **Leanne M. Redman:** Writing – review & editing, Writing – original draft, Conceptualization.

used to diagnose obesity. Prevalence rates of obesity have nearly tripled globally since 1975 (1), which contributes to increasing morbidity and mortality for many noncommunicable diseases, including cardiovascular diseases, diabetes, and at least 13 types of cancer. The United States (US) has one of the highest obesity prevalence rates in the world, with the latest estimates suggesting that >42% of the adult population is affected (2). Furthermore, the occurrence of adults who are diagnosed with class II (35–39.9 kg/m²) and class III (severe obesity; ≥40.0 kg/m²) obesity has doubled in recent years, with class III obesity projected to increase at the highest rate in the coming decade (3).

The health of females throughout their reproductive years is critical for optimizing fertility, maternal and infant outcomes associated with pregnancy and childbirth (4), and lifelong health risks for women and children (5, 6). Indeed, all-cause mortality and cardiovascular events are increased in the offspring of mothers with obesity (7). In US women of reproductive age (18–40 years), obesity rates increased fourfold from 7.4% in 1976 to 27.5% in 2014 (8); to date, >40% of women 20–39 years are estimated to be living with obesity (2). Maternal prepregnancy obesity has been associated with an increased risk of obesity and overweight at early ages in offspring conceived with fertility treatments (9). Maternal BMI may also affect the risk of birth defects, such as spina bifida, and intellectual disabilities in offspring (9, 10). Because obesity-related adverse chronic health and behavioral risk factors during preconception can have detrimental effects on fertility, pregnancy, and maternal and offspring health outcomes (6), the prevention and management of obesity in females of reproductive age is critically important.

Obesity pathophysiology

There are 2 prevailing models of obesity pathogenesis: the energy balance model and the carbohydrate-insulin model. Although each agrees that there is a fundamental metabolic reprogramming of energy partitioning that favors excess energy deposition in fat, they differ in their explanation of pathways leading to chronic energy imbalance. The energy balance model suggests overeating coupled with physical inactivity and untoward metabolic adaptations to the obesogenic environment drive excess adiposity, although the carbohydrate-insulin model theorizes that increased adiposity drives overeating (11).

The energy balance model represents the most traditional and widely accepted concept for obesity, which states that obesity pathogenesis ensues from a chronic energy imbalance, whereby a chronic overconsumption of dietary energy that is not matched by basal metabolic rate and physical activity energy expenditure and excess energy is deposited in adipose tissue depots, leading to body mass expansion (Fig. 1) (12). In females of reproductive age, the relationship between energy balance and reproduction is tightly coupled. Reproductive processes are energetically expensive. Compared with males, females have a higher percent body fat mass and a greater tendency to store excess nutrients during surplus energy intake as a protective mechanism against undernutrition (13). Energy and nutritional status have direct effects on sex hormone production, ovulation, menstrual cycle regularity, fertility, conception, and early embryonic development (14).

The carbohydrate-insulin model of obesity, which is newly described, suggests that the chronic expansion of adipose tissue stores is driven by increased consumption of dietary

carbohydrates because of their affinity for insulin secretion. Excess consumption of dietary carbohydrates and prolonged increases in circulating glucose stimulate a prolonged secretion of insulin, which directs dietary energy to fat storage in adipose tissue as opposed to oxidation by metabolically active tissues (11). In turn, cells are postulated to be “starved” of energy, driving an increase in dietary energy intake. In females of reproductive age, insulin is a key metabolic hormone with direct actions on the gonadotropin-releasing hormone neuronal network, which regulates reproduction (13, 15). Hence, variations in insulin secretion patterns influenced by diet composition and feeding frequency could subsequently impact energy partitioning to a greater extent than positive energy balance.

Increased intake of all 3 macronutrients (i.e., carbohydrates, fat, and protein) has been associated with insulin resistance, obesity, and metabolic diseases (16). A recent exploration regarding the composition of fat intake (e.g., low polyunsaturated fatty acids) has demonstrated that the excessive formation of free fatty acids and inappropriate lipid deposition in organs other than adipose tissue, also known as lipotoxicity, underlies the development of oxidative stress and mitochondrial dysfunction (17, 18). This most notably occurs through insulin resistance (18). Yet it is unclear whether increased consumption of one macronutrient is more detrimental than another regarding energy balance regulation.

Body weight is a central regulation. The brain relies on adipose- and gut-derived hormones, such as leptin, insulin, and ghrelin, to regulate appetite, satiety, energy expenditure, and hence energy stores (13, 19). It is also well established that these metabolic hormones are linked to fat stores, and they impact the control of body weight through long-term effects on energy balance (19). For example, the adipose tissue-derived hormone leptin regulates food intake and energy expenditure through hormonal signaling to the hypothalamus via a negative feedback loop to control adipose tissue mass (20). Individuals with obesity are often leptin-resistant, disrupting the normal central regulation of food intake and adipose tissue mass, particularly as the lower leptin levels decrease energy expenditure and inhibit appetite regulation (19, 21–23). Additionally, glucagon-like peptide-1 (GLP-1), a gut hormone vital to glucose homeostasis, acts through the GLP-1 receptor, which is expressed in the brain, gut, and pancreas (24, 25). Glucagon-like peptide-1 reduces blood glucose levels by stimulating insulin secretion, which is important in those who are insulin-resistant (26, 27). It also acts by inhibiting glucagon secretion, which reduces endogenous glucose production, subsequently reducing the drive for food intake and slowing gastric emptying. This interaction between the gut and brain is possibly mediated by neuropeptide Y, agouti-related peptide, and proopiomelanocortin neurons (24, 26). Obesity interferes with gut hormones (e.g., GLP-1) ability to secrete peptides (e.g., agouti-related peptide and peptide tyrosine), which are critical for the homeostatic control of body mass through regulating food intake (brain) and energy expenditure (metabolism) (24, 28).

Obesity phenotypes

With BMI arguably an imperfect measure of obesity (29), a wide heterogeneity of obesity-related comorbidities exists across individuals. Obesity that is present in conjunction with cardiometabolic complications (e.g., hypertension and hyperlipidemia) and metabolic disorders (e.g., prediabetes and diabetes) is linked with worsened disease progression and

increased mortality (30, 31). When obesity presents in the absence of other comorbidities, it has been defined as metabolically healthy obesity. These individuals are likely in a transient state, with more chronic disease morbidities to develop eventually (32). Consequently, metabolically unhealthy obesity is defined as obesity plus risk factors for cardiometabolic disease (33). Although there is no clinical practice guideline for defining obesity phenotypes, the 4 diagnostic criteria for metabolic syndrome are most commonly applied, with the presence of ≥ 2 of the 4 diagnostic criteria (insulin resistance, impaired fasting glucose and/or tolerance, dyslipidemia, and hypertension) conferring a positive diagnosis of metabolically unhealthy obesity (33, 34). It is important to emphasize that although each mechanism of obesity development leads to chronic energy imbalance, none is the result of a person being lazy or lacking willpower. As discussed in detail below, evaluating when a patient of reproductive age has obesity plus other signs of cardiometabolic disease is important to understanding their reproductive abilities, future pregnancy complications, and long-term gynecological and general health (35, 36). Finally, evaluating obesity beyond weight can also reduce stigmatization often experienced by patients and allow for more holistic and personalized clinical management (37).

Beyond other existing comorbidities, obesity phenotypes have been also proposed for 1 of 4 different underlying pathophysiologies, and these can also be diagnosed clinically: hungry brain (defined by abnormal satiation), emotional hunger (defined by hedonic eating), hungry gut (defined by abnormal satiety), and slow burn (decreased metabolic rate) (38). To identify these pathophysiologies, clinical tests are utilized. Homeostatic eating behavior can be divided into 3 stages: hunger (desire to eat assessed using a visual analogue scale), satiation (calories consumed to reach fullness and terminate a meal assessed using an ad libitum buffet), and satiety (duration of fullness or return of hunger assessed using a visual analogue scale). Hedonic eating behavior can be assessed using the hospital anxiety and depression scale and a three-factor eating questionnaire, whereas energy expenditure can be evaluated using indirect calorimetry (38). In a 12-month pragmatic weight management trial with 450 adults, 32% of patients presented with a hungry brain, 32% with a hungry gut, 21% with emotional hunger, and 21% with slow burn (38). Understanding the underlying pathophysiology of each patient's obesity has allowed for a more personalized approach to care, particularly with medications. For example, individuals with a hungry brain phenotype may be more responsive to phentermine-topiramate, although those with a hungry gut may respond best to liraglutide, a GLP-1 agonist (38). Using these 4 clinical phenotypes in future research studies may help elucidate underlying mechanisms for successful interventions (38), thereby aiding in health promotion for at-risk populations such as reproductive-aged women.

OBESITY AND FERTILITY

Hypothalamic-pituitary-ovarian axis

The impact of obesity on reproduction is both complex and multifactorial. Clinically, females with obesity frequently experience irregular menses, with systematic observations of obesity-associated menstrual disturbances described for >70 years (39–44). The probability of menstrual irregularities corresponds with the degree of obesity, with an increasing

risk for oligo- and amenorrhea associated with increasing BMI (41, 45). However, as discussed above, there are significant limitations to using BMI as the sole metric of pathologic adiposity. Studies have shown that outside of BMI, abdominal adiposity and subcutaneous abdominal adiposity are most significantly associated with anovulation (46, 47). Furthermore, the age of obesity onset also seems to impact the risk of anovulation. For example, overweight and obesity at age 7 and 23 years both independently increase the risk of menstrual dysfunction and subfertility at age 33 years (42).

Arguably the most well-established mechanism of obesity-induced menstrual dysfunction involves interruption of the hypothalamic-pituitary-ovarian (HPO) axis, resulting in anovulation. This is most evident in patients with polycystic ovary syndrome (PCOS), where excess adipose deposition, especially in central depots, results in hyperinsulinemia, increased androgen production, decreased sex-hormone binding globulin levels, and increased peripheral aromatization, all of which can have direct and/or indirect actions on the HPO axis (48, 49). A full review of PCOS and the associated metabolic derangements associated with obesity is outside of the scope of this review and has been reviewed recently elsewhere (50).

Among eumenorrheic females without PCOS, obesity often results in a relative state of hypogonadotropic hypogonadism (43, 51, 52). Studies of eumenorrheic females with overweight and obesity compared with those with normal weight, report decreased serum follicular phase levels of follicle-stimulating hormone (FSH), luteinizing hormone (LH), and inhibin B (51, 52). Recent work aiming to further understand the mechanisms of relative hypogonadism, assessed by simulating the pathologic milieu of obesity, found that hyperinsulinemia combined with hyperlipidemia acutely suppressed serum levels of both FSH and LH (53). Using a gonadotropin-releasing hormone agonist to suppress the HPO-axis followed by intravenous administration of FSH, a recent study found that there is likely an effect of obesity beyond acute pituitary suppression (54). Although equivalent serum levels of FSH and pharmacodynamics were achieved in women with and without obesity, women with obesity exhibited both decreased baseline levels of estradiol and inhibin B and a significantly attenuated response to FSH levels. This suggests that beyond the obesity-induced pituitary effects described above, there is likely a direct ovarian effect (54). Urinary levels of pregnenediol glucuronide are also reduced among women with obesity (52).

Ovaries

Obesity has a direct effect on the ovary and oocyte. Preclinical models of diet-induced obesity demonstrate primordial follicle depletion, follicle apoptosis, decreased mature follicle size and number, increased oocyte meiotic aneuploidy, and altered oocyte mitochondria (55–59). Although diet reversal improves the circulating metabolic milieu, meiotic and mitochondrial abnormalities persist, suggesting that obesity may induce an irreversible insult to the ovary (60).

Similarly, human studies demonstrate morphological oocyte abnormalities and decreased oocyte competence with obesity. Oocytes from females with obesity undergoing in vitro fertilization (IVF) treatment are often smaller compared with their normal weight counterparts, are less likely to complete development postfertilization, and in 1 study, were

noted to have granular cytoplasm (61–63). In addition, obesity is associated with meiotic spindle abnormalities and fewer mature oocytes (64–66). Additionally, resulting blastocysts demonstrate altered metabolism, with reduced glucose consumption and increased levels of triglycerides (62). Moreover, the follicular fluid of females with obesity demonstrates significantly altered adipokine composition, inflammatory markers (e.g., C-reactive protein), and free fatty acids (67–70). Circulating inflammatory markers have been also shown to correlate with intrafollicular levels of free fatty acids and triglycerides (67).

The increased systemic leptin present with obesity is similarly evident in the follicular fluid of women with obesity, who have higher levels of intrafollicular leptin compared with women of normal weight (67, 71, 72). In vitro studies utilizing human granulosa cells obtained from women undergoing IVF treatment demonstrate that leptin can directly inhibit basal and FSH-stimulated estradiol and progesterone levels (73). Leptin has been also shown to directly suppress both antimüllerian hormone (AMH), a key hormone in developing fetal sex organs in vitro, and its receptor AMH receptor II's messenger ribonucleic acid levels in human granulosa cells (73).

There is some controversy regarding whether or not BMI is consistently associated with serum AMH levels, especially among women with regular menses (74). However, several recent large-scale studies demonstrated significantly lower AMH levels among women with and without PCOS (75–77). In the appraisal of body content study involving 1,706 eumenorrheic women without PCOS, obesity on the basis of either percent body fat or BMI was associated with lower serum AMH levels (76). Similarly, in a cohort of reproductive-aged African-American women ($n = 1,654$), AMH concentrations were 23.7% lower in women with obesity compared with women with normal weight (75). In addition, this study suggested a possible cumulative effect of obesity: women with obesity present since age 18 years had AMH levels that were significantly lower than women with normal weight at age 18 years but who currently had obesity. Finally, a study using samples from the pregnancy in PCOS (PPCOS) II and ovarian aging studies (640 women with PCOS and 921 ovulatory control women) found that increasing BMI and waist circumference were associated with a reduction in AMH levels as well as AMH levels and antral follicle count (a marker of per-follicle AMH) (77).

Uterus

Obesity has a well-established association with endometrial cancer (78). Indeed, it not only increases the risk of endometrial hyperplasia and endometrial cancer but also increases the risk of mortality and recurrence posttreatment (79). Given the clear impact of hyperestrogenism on the endometrium, the impact of obesity on the endometrium because it relates to implantation, fertility, and infertility has come into view.

Clinically, models to assess the isolated impact of obesity on the endometrium use a donor oocyte model where embryos created from normal-weight oocyte donors are transferred into women with and without obesity. Although a meta-analysis including 4,758 women demonstrated no association between recipient obesity and the chance of pregnancy after IVF treatment (80), these results were recently contradicted by 2 large-scale studies that reported an independent endometrial effect associated with obesity (81, 82). Worthy of

mention, a study using Society for Assisted Reproductive Technology data to assess live birth (LB) rates from frozen embryo cycles using preimplantation genetic testing for aneuploidy embryos and including 77,018 preimplantation genetic testing for aneuploidy cycles (6,266 of which were donor oocyte recipients) found a significant relationship between BMI and LB. This relationship was observed in both those using autologous and donor oocytes and was especially evident in patients with a BMI (≥ 40 kg/m²) (82). Notably, the presented studies utilized BMI as the sole metric of obesity, which, as previously discussed, does not necessarily capture the complexity of obesity and may explain some of the contradictory findings above. Additionally, many practices have BMI restrictions, and thus higher levels of obesity may not be reflected (83).

Preclinical models of diet-induced obesity demonstrate an independent endometrial effect of obesity, with evidence of a decreased decidualization response to hormonal stimulation and decreased numbers of implantation sites (84). A high-fat diet has been shown to induce significant alterations to gene expression in mouse endometrial epithelia and stroma, with altered expression of genes related to innate immunity, leukocyte chemotaxis, and circadian rhythms (85). Studies of human endometrium report altered proteomic and transcriptomic profiles with obesity (86–88). It has been also suggested that obesity may displace the window of implantation, especially in the setting of stage II (BMI 35–39.9 kg/m²) and III obesity (BMI ≥ 40 kg/m²) (89). Finally, both leptin and adiponectin are expressed in the human endometrium, with altered protein and/or receptor expression found in women with recurrent implantation failure (90). In vitro studies using human endometrial cell lines suggest that leptin may also have a direct role in endometrial epithelial remodeling and endometrial receptivity during implantation (91, 92).

FERTILITY TREATMENTS

As reviewed above, obesity has a complicated and significant impact on all aspects of the HPO axis and also directly impacts the ovaries and uterus. Clinically, this culminates in longer-time-to pregnancy (even among eumenorrheic females) and increased risk for infertility (93–97).

Among females with anovulatory infertility, BMI was found to be an independent predictor of a decreased chance of LB with clomiphene citrate ovulation induction (98). In the PPCOS I study, participants with a BMI < 30 kg/m² had significantly higher rates of LB compared with those with a BMI > 30 kg/m² (99). Similarly, BMI remained a significant factor in rates of LB in PPCOS II (100).

Females with obesity require higher doses of gonadotropins, have lower estradiol levels, and are at increased risk for cycle cancellation (101–106). Despite similar percentages of euploid embryos (107, 108), most studies evaluating pregnancy and LB outcomes that include fresh embryo transfers (ETs) demonstrate an association with BMI. For example, using data from the Center for Disease Control and Prevention's National ART Surveillance System and 494,097 treatment cycles, females with obesity had a statistically decreased likelihood of both intrauterine pregnancy and LB (109). This same trend was recently confirmed in a non-US (Swedish) population using national data (110). Although it was suggested that the

impact of obesity on LB may be less evident among women undergoing frozen ETs (111), 2 recent large-scale studies utilizing Society for Assisted Reproductive Technology data and only examining frozen ET cycles of euploid embryos did demonstrate an independent effect of BMI (82, 112) and women with a BMI >40 kg/m² had a 27% lower probability of LB compared with the normal weight reference group (82). It is extremely important to note, however, that although there is a consistent impact of obesity on IVF treatment outcomes, this effect is significantly modified by age (113, 114). Thus, although obesity may decrease the chance of LB, the powerful effect of age must always be considered.

OBSTETRICAL OUTCOMES

Miscarriage

A number of studies demonstrate an association between maternal obesity and an increased risk of miscarriage (115–120). This increased risk is apparent after pregnancies conceived spontaneously (115, 117) and those achieved after fertility treatments (116–120). It was also demonstrated that women with obesity have an increased frequency of euploid miscarriage compared with women with normal weight, suggesting nonchromosomal etiologies of pregnancy loss in this population (121, 122). Obesity is also an independent risk factor among populations of women with recurrent pregnancy loss and increases the risk of future pregnancy loss (122–124). As noted in the 2021 American Society of Reproductive Medicine committee opinion on obesity, the adjusted odds ratio for most studies describing these associations is modest and ranges between 1.2 and 1.9 (93). Thus, although a relationship between BMI and miscarriages is consistently demonstrated, there is a risk of confounding factors that are likely influenced by the heterogenous metabolic, inflammatory, and reproductive profiles associated with excess adiposity, as well as how obesity was defined in each study.

Adverse pregnancy outcomes

Females of reproductive age have an increased risk of longitudinal weight gain and consequent obesity (125). Longitudinal studies report that females gain, on average, up to 0.7 kg per year, with the greatest rates of weight gain occurring in females aged 18–50 years (125). Existing obesity-related cardiometabolic comorbidities before conception often digress into obesity-related adverse pregnancy outcomes, such as gestational diabetes and preeclampsia (5). In epidemiological studies of US national data, the occurrence of gestational diabetes increased exponentially from 3.6% in females with a normal weight BMI to 8.8%–13.9% in females with a BMI ≥ 30.0 kg/m² (126). Women with obesity who develop gestational diabetes are more likely to develop type 2 diabetes postpartum than those without gestational diabetes (127). Epidemiological data also demonstrate that hypertensive disorders affect approximately 6% of pregnancies in the US with a positive relationship with increasing BMI (5). Similarly, women who had hypertensive disorders related to obesity during pregnancy had a higher prevalence of chronic hypertension later in life compared with those who did not (49.6% vs. 21.5%, respectively) (128). Taken together, obesity-related adverse pregnancy outcomes can lead to a higher probability that women will develop metabolically unhealthy obesity postpartum, even if they were considered metabolically healthy before conception.

Weight gain in pregnancy compounds the effect of pregravid obesity on adverse pregnancy outcomes (5). Only approximately one-third of women gain the Institute of Medicine's recommended amount of weight during pregnancy, with most women gaining weight outside the recommendations (21% too little; 48% too much) (129). Excess gestational weight gain in pregnant women with obesity, defined as >9 kg, may further exasperate adverse pregnancy outcomes (e.g., gestational diabetes mellitus, large for gestational age infants, nonelective cesarean delivery, and others.), leading to complications during gestation and delivery (5). The pervasiveness of obesity-related adverse pregnancy outcomes can affect subsequent pregnancies as well, indicating the need for effective weight management before conception, during gestation, and postpartum.

Implications of preconception weight loss

Numerous studies suggest that weight loss during prepregnancy and during gestation can significantly reduce the risk of obesity-related adverse pregnancy outcomes. Observational studies that followed 415,605 women for approximately 10.6 years (range, 2–22 years) demonstrated that weight loss with lifestyle interventions was associated with a reduced risk of hypertensive disorders during pregnancy and postpartum (130). Other studies observed a positive effect of weight loss (–4.54 kg) between pregnancies on the risk of gestational diabetes during the subsequent pregnancy, highlighting the benefits of weight loss preconception (30). Combining lifestyle interventions with other antiobesity treatments, such as bariatric surgery or weight loss medications, can improve health parameters before conception. Indeed, an evaluation of over 2.8 million women (of whom 8,364 have undergone surgery) found that bariatric surgery before conception was significantly associated with reductions in gestational diabetes, hypertensive disorders, and cesarean delivery during pregnancy (131). Bariatric surgery can also improve ovulation, reduce the risk of miscarriage, and improve perinatal outcomes (132). The major untoward effect of bariatric surgery appears to be the increased risk for small gestational-age infants because of weight loss throughout pregnancy, yet this risk can be minimized by waiting the recommended 12 months after bariatric surgery to attempt to conceive and by focusing on good nutrition intake and supplementation during pregnancy. It is important to note that although weight loss may improve obstetrical outcomes, randomized controlled trials do not demonstrate improvements in LB in the setting of preconception weight loss, especially among ovulatory individuals (discussed in additional detail in the accompanying review) (93, 133). However, treatments for obesity before conception in women of reproductive age can promote healthier pregnancies, reduce maternal and neonatal adverse outcomes, and have an intergenerational effect, potentially contributing to a lower incidence of childhood obesity (134).

RESEARCH GAPS AND CLINICAL NEEDS

Precision medicine

Despite the fact that over 40% of reproductive-aged women in the US are living with obesity, the current US obesity guidelines do not consider prepregnancy weight or the prevalence of obesity-related adverse pregnancy outcomes in the decision to escalate obesity therapies (2, 134). This lack of guidelines for women of reproductive age

signifies a potentially missed opportunity to initiate meaningful weight-loss interventions preconception and postpartum to attenuate the effects of obesity on maternal health. Precision medicine is an emerging innovative approach to managing obesity that considers individual differences in genes (i.e., polygenic risk score and single nucleotide polymorphisms), environments, and lifestyles. Women of reproductive age represent a population that would greatly benefit from precision medicine, particularly in relation to nutrition and lifestyle (135). Identifying specific diet and physical activity components for preventing weight gain, excess adiposity, and obesity is important for creating a strong evidence base for tailoring intervention strategies aimed at improving health in reproductive-aged women with obesity. Studying the newly proposed pathologies of obesity (38) in a cohort of reproductive-aged females in conjunction with a neuroendocrine workup of their reproductive function would fill a major knowledge gap.

Tools to identify high-risk patients

Because obesity is considered a medical risk factor for a high-risk pregnancy, early prenatal care is recommended to evaluate the cardiometabolic profile and other possible risk factors often associated with obesity. Laboratory tests for assessment of nutritional status and metabolic profiles are recommended in addition to dietary assessments using a food frequency questionnaire (136). Personalized nutrition interventions focused on reduced caloric intake and addressing nutritional needs can aid in managing maternal weight preconception. Common tools utilized to evaluate high-risk women during pregnancy are blood and urine tests, ultrasonography, frequent blood pressure measurements, and regular follow-ups with medical staff. Additionally, the US Preventive Service Task Force issued a grade B recommendation for using behavioral interventions before conception and during pregnancy to manage obesity-related adverse pregnancy outcomes via gestational weight gain (137). Effective interventions for weight management focus on individual nutrition, physical activity, and/or lifestyle change delivered by clinicians, registered dietitians, qualified fitness specialists, physiotherapists, and health coaches across different settings (e.g., local community fitness centers) (137). However, in the absence of significant weight loss with the initiation of comprehensive lifestyle interventions, escalation of therapy should be considered.

Effective interventions and therapeutics for weight management before pregnancy

The use of obesity phenotypes may aid in addressing the limitations of BMI and the heterogeneity in obesity-related to varied weight loss in response to obesity interventions, such as diets, medications, devices, and surgery. First, evaluating when individuals have metabolically healthy or metabolically unhealthy obesity is a simple and available method that can guide practitioners in providing recommendations for personalized care (e.g., a low-fat diet for dyslipidemia). Second, assessing obesity pathophysiology may further improve obesity care for females of reproductive age. Assessing homeostatic eating behavior (hunger, satiation, and satiety) can be completed through widely available questionnaires, although hedonic eating behavior and energy expenditure assessment require measures that may not be available clinically. Although differentiating between the phenotypes can provide a more in-depth assessment of potential mechanisms to guide individualized therapy for obesity (38), using accessible measures for homeostatic eating behavior can be a key

tool clinicians can add to their obesity care. A multidisciplinary team, such as registered dietitians and behavioral specialists, may be an acceptable way to evaluate obesity in a more personalized manner. Understanding the key determinants of an individual's eating behavior and energy intake can be addressed with behavioral counseling in addition to therapeutic interventions such as weight loss medications and bariatric surgery. Before conception, antiobesity medications are approved for women of reproductive age. A prepregnancy weight loss goal of 5%–7% is a reasonable target and approximates the additional total body weight loss achieved by adjunct antiobesity medications (134). The use of medications may be an effective therapy to treat obesity that does not have as many long-term risk factors compared with bariatric surgery, but the use of medications should be paired with lifestyle interventions to prevent weight gain during pregnancy and postpartum. This is particularly important for females using medications before conception to ensure proper lifestyle skills are developed for managing weight during pregnancy and in the postpartum time period. Because obesity is multifactorial, practitioners should be aware that finding an effective solution will also be multidimensional and probably require lifelong approaches.

CONCLUSION

The health of nonpregnant women during their reproductive years is critical for optimizing fertility success and reducing maternal and infant mortality and pregnancy complications. Females of reproductive age with obesity may experience challenges to their reproductive health via direct effects on the HPO axis, the ovary and oocyte, and the endometrium. Thus, understanding the complex pathophysiology of obesity and utilizing phenotypes that go beyond the limitations of BMI is imperative for providing personalized clinical care for females of reproductive age.

Declaration of Interests

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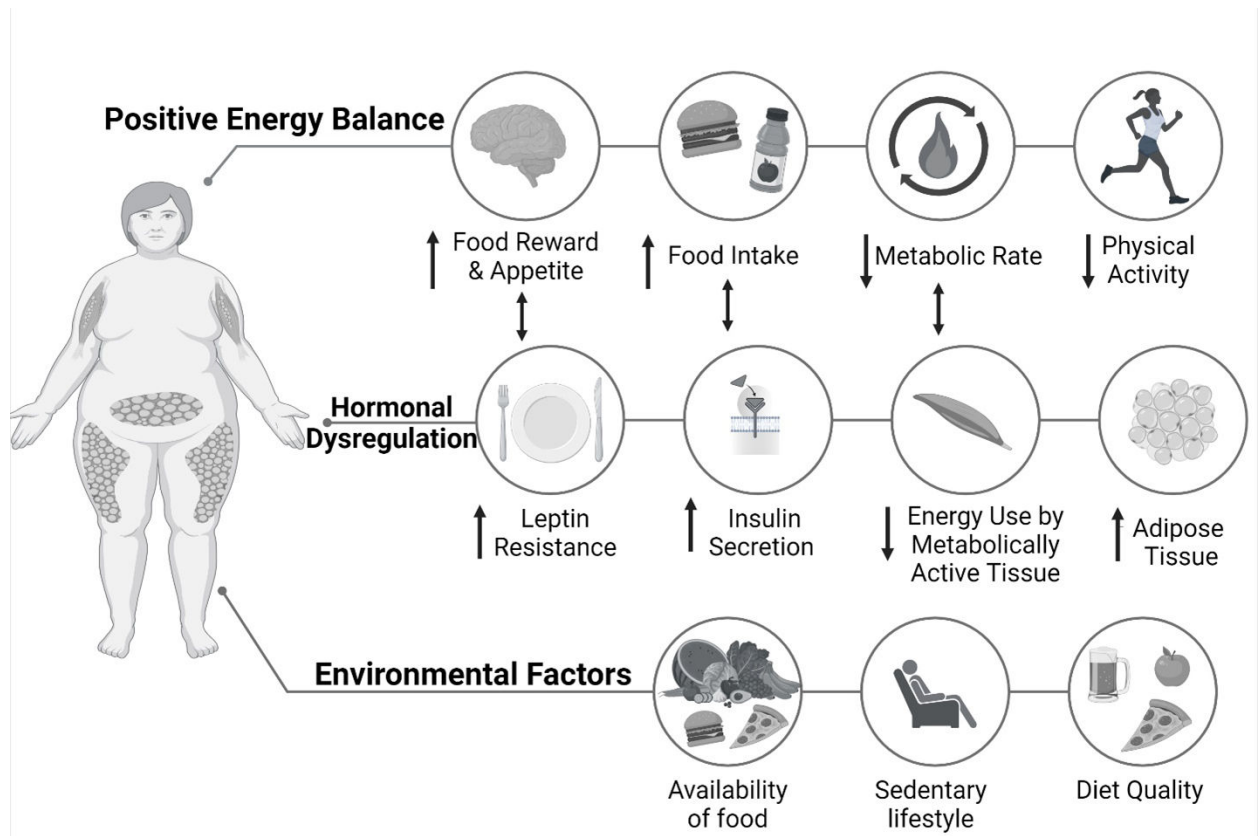


FIGURE 1. Overview of the pathophysiology of obesity.
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